Computing Infrastructures

Riassunto completo del corso

Politecnico di Milano

A.A. 2025-2026, II Semestre

Contents

1 Prefazione **2**
- 1.1 Disclaimer . . . 2
- 1.2 Keywords . . . 2

2 Datacenters **2**
- 2.1 Introduction . . . 2
- 2.2 Hardware Infrastructures . . . 3
  - 2.2.1 Data Centers . . . 4
  - 2.2.2 Edge, Embedded and IoT Devices . . . 4

3 Data Center as a Computer **4**
- 3.1 From Data Centers to Warehouse-Scale Computers... . . . 5
- 3.2 ...and Back . . . 6
- 3.3 Areas and Regions . . . 6
  - 3.3.1 Edge DataCenters . . . 7
  - 3.3.2 Availability and Uptime . . . 8
- 3.4 Architectural Overview of WSCs . . . 8

4 Servers **9**
- 4.1 Accelerators and ML Era . . . 10
- 4.2 Tensor Processing Units . . . 11

1 Prefazione

1.1 Disclaimer

I contenuti di questo PDF sono stati compilati con il solo scopo di avere una risorsa esaustiva e facilmente consultabile degli argomenti trattati nel corso di Computing Infrastructures sostenuto dal professore Palermo: non sono in alcun modo materiale sostitutivo alle lezioni, che frequentate in presenza mi hanno permesso di capire a fondo la materia. Sconsiglio di usare esclusivamente suddetto documento per cercare di prepararsi all'esame, supporto invece l'uso di questa dispensa per fare il punto della situazione durante il periodo di studio.

1.2 Keywords

- **DC** - Data Center
- **IoT** - Internet of Things
- **AI** - Artificial Intelligence
- **IaaS** - Infrastructure as a Service
- **PaaS** - Platform as a Service
- **SaaS** - Software as a Service
- **LAN** - Local Area Network

2 Datacenters

2.1 Introduction

Datacenter

A Datacenter is a building whose primary purpose is to house a computer room and its support areas, so that services can run continuously and efficiently

There exist lots of different types of datacenters:

- **Enterprise Data Centers** - facilities operated and controlled by the company that provides the service. Main problem: expensive to maintain (to build or not to build dilemma) and can shorten the possibility to generate money by the business.
- **Colocation Data Centers** - larger facilities built to accomodate different businesses and can personalize individual IT parts for individual businesses. Main problem: IT purchases are on your own and restrictions on access and maintenance.
- **Managed Hosting Platforms** - all facilities are managed by third parties, still a single-tenant solution. Main problem: higher operational costs and possibility of customer lock in problem.

In addition to the above solutions, there are also **Cloud Data Centers**, which are the most common solution in post-pandemic times. Those solutions are **IaaS**, **PaaS** and **SaaS**. A clear representation is shown in the figure below.

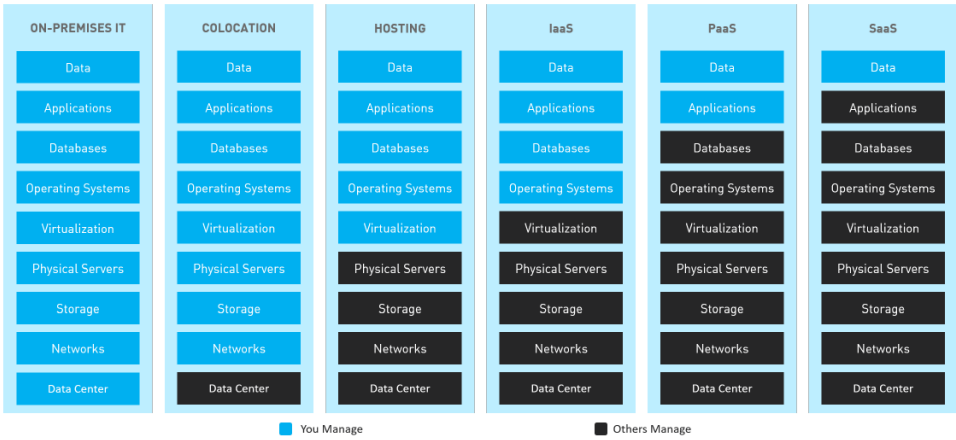

Figure 1: Cloud Data Centers

One thing that stands out is that cloud solutions also contain virtualization managed by the provider (you don't know which operating system is running, you are inside an additional abstraction layer).

2.2 Hardware Infrastructures

Let's start with the basic definition of a computing infrastructure.

Computing Infrastructure

A **Computing Infrastructure** is a tecnological infrastructure that provides hardware and software for computation to other systems and services.

Computing infrastructures can be categorized in a pyramid of three layers:

- **IoT and AI-enabled Edge Sensors** - where data acquisition and processing happens at the edge of the network.
- **Edge Servers** - on-premises hardware resources that perform more intensive processing tasks.
- **Data Centers and Cloud** - virtualized computing and capacity on demand.

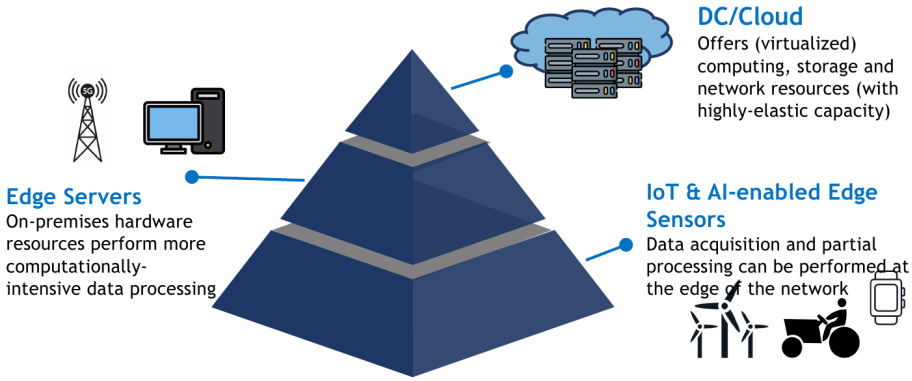

Figure 2: Computing Infrastructure Pyramid

It's obvious that the bottom layer has less memory and computational power, but it is also more responsive and with low latency; the upper layers offer more resources to work with. To categorize all the components of a system it is used what is called the **Computing Continuum**.

2.2.1 Data Centers

Figure 3: Data Centers

Usually, data centers contain servers for **processing**, **storage** and **communication**. They have lower IT costs, high performance, "unlimited" storage capacity, universal data access and have device independence. On the other hand they need a constant internet connection, they do not work well with low speed connections, have high power consumption and cooling costs. Most importantly they add **latency in taking decisions**.

2.2.2 Edge, Embedded and IoT Devices

- **IoT Devices** - useful for their pervasiveness, wireless connection and low costs, sometimes powered by batteries, but with limited resources, security issues and difficult to program.
- **Embedded PCs** - support pervasive computing, with more resources and abstraction layers (programmed as PCs), but with higher costs and power consumption.
- **Edge/Fog Computing Systems** - have high computational capacities, better in terms of privacy, have reduced latency in making decisions, but with higher costs of power and with cloud connection requirements.

What is Edge and what is Fog

In Edge Computing, data is processed at the edge of the network, whereas in Fog Computing data is moved into a LAN before going into a DC.

3 Data Center as a Computer

In Data Center history we witnessed a shift towards smaller devices used as client actors inside our systems: with the advent of cloud computing and mobile devices, portability and scalability

are being able to further develop inside our applications. Also, it's easier to update the data inside a shared and cloud connected database instead to update millions of devices around the world, users have ease of management (with little configurations needed) and many application services can run at a low cost per user: it is worth noting that many application require the strength, power and availability of a full datacenter to perform some specific workloads (see search engines).

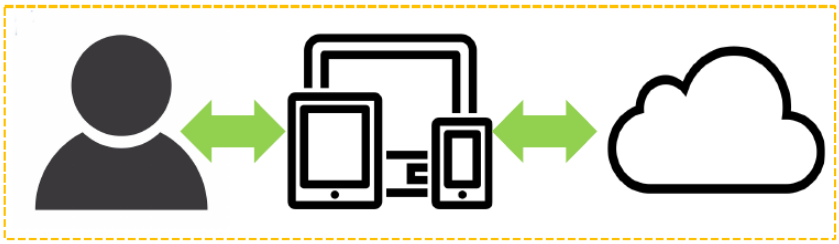

Figure 4: Distributed Software

3.1 From Data Centers to Warehouse-Scale Computers...

This trend created a new class of computing systems that needed to emerge from regular datacenters whose main characteristic was to have lots of power to host a large number of small-medium sized applications, each one running on dedicated hardware and isolated with software for multiple organizations. **WSC** are made for single organizations, where the hardware is relatively homogeneous and applications have a common system manager layer and talk to each other: it can be considered as a large cluster of servers that needs to be considered as a single computing unit. It is meant to run a smaller number of very large applications.

Warehouse-Scale Computers

WSC are large scale datacenters accomunated by homogeneous hardware and are to be considered a single computational unit made by a single organization for the developement of multiple large-scale applications.

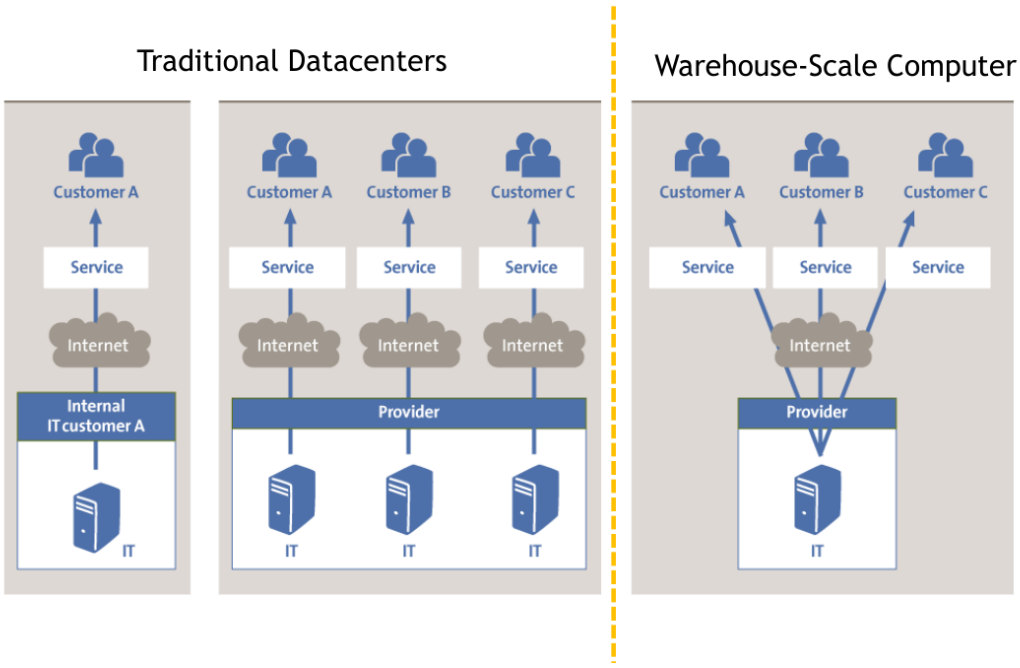

Figure 5: Traditional Data Center vs WSC

3.2 ...and Back

Right now, Amazon, Microsoft and Google are all using this model for their public clouds computing systems, but in a more traditional-oriented way: since such public clouds use lots of little applications, they are operated like a datacenter thanks to virtualization inside the system. WSC implements the traditional paradigm thanks to Virtual Machines.

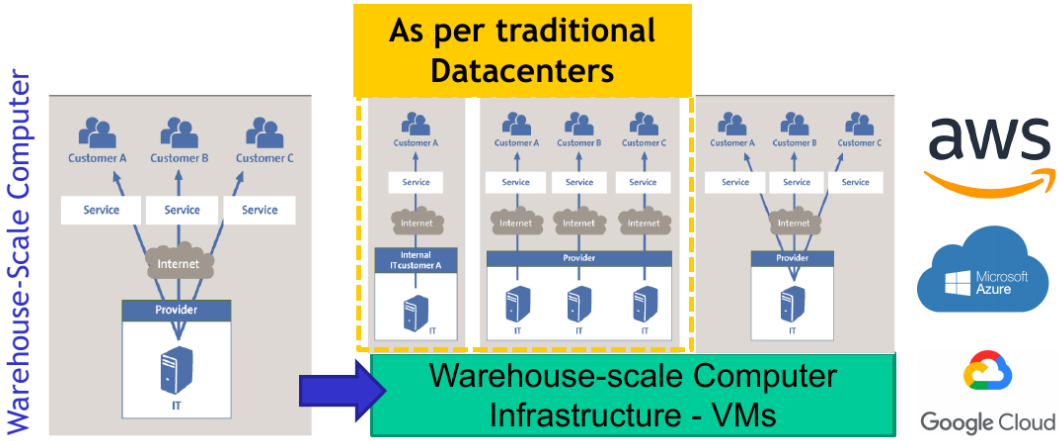

Figure 6: WSC with Virtualization

DataCenter Replicas

Even if single requests are typically processed by a single datacenter, it is good practice to replicate the same datacenter to reduce latency and improve throughput.

3.3 Areas and Regions

To be sure that all the users are served it is possible to use different approaches:

- **Geographic Areas (GAs)** - we can use geopolitical boundaries to better distribute the loads and to grant data residency: it is mandatory that in the same GA there are at least two computing regions
- **Computer Regions (CRs)** - a finer grain of location: it is confined by latency (¡2ms), distance (¡100 miles apart) and disaster recovery.
- **Availability Zones (AZs)** - a finer grain of location within a single **CR**: multiple DC in a single CR allows to run mission critical applications, syncronicity of data is granted by a quorum of minimum three data centers.

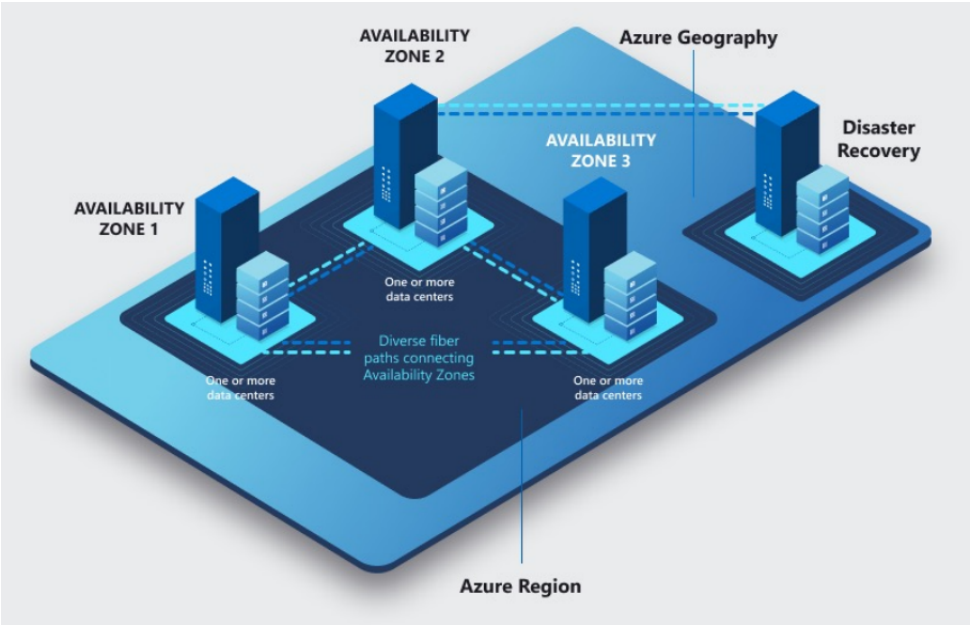

Figure 7: Azure Architecture inside a **GA**

3.3.1 Edge DataCenters

To satisfy latency and to grant coverage in all areas, sometimes companies use **Edge DataCenters**, smaller datacenters made exclusively to filter requests that were going to the main datacenters. Sometimes, companies prefer to have many Edge DataCenters that serve a single main DataCenter.

Figure 8: Map of Amazon AWS Edge DataCenters with relative Main DataCenters

3.3.2 Availability and Uptime

Usually Uptime is pretty high among those type of datacenters:

- **Single Instance VMs** - 99.90%
- **Availability Sets** - 99.95%
- **Availability Zones** - 99.99%

3.4 Architectural Overview of WSCs

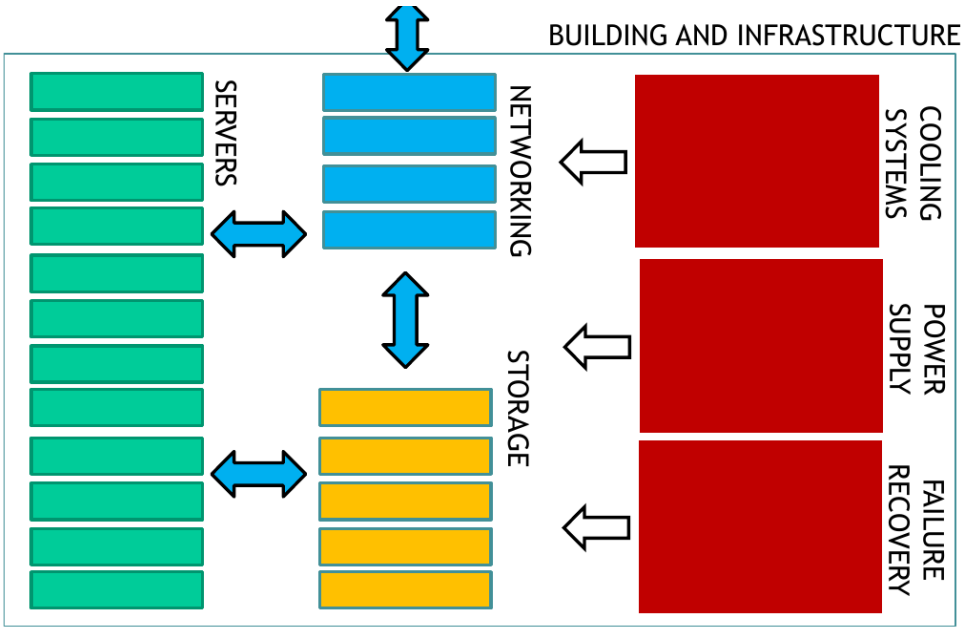

Figure 9: WSC General Architecture

- **Servers and Rack** - the main processing equipement, hosted inside shelves as basic building blocks of DC and WSC, connected by hirarchies of network.

- **Storage** - the backbone of the system, composed in different ways depending on the necessities (**DAS**, **NAS**, **SAN**, **RAID**).
- **Networking** - also called **DCN** (DataCenter Network), connects every component of the system with switches, routers, load balancers and cables.
- **Power, Cooling and Failure Recovery** - all the necessary to maintain the whole infrastructure solid and working.

4 Servers

We can now focus on the single nodes of our systems: **servers**.

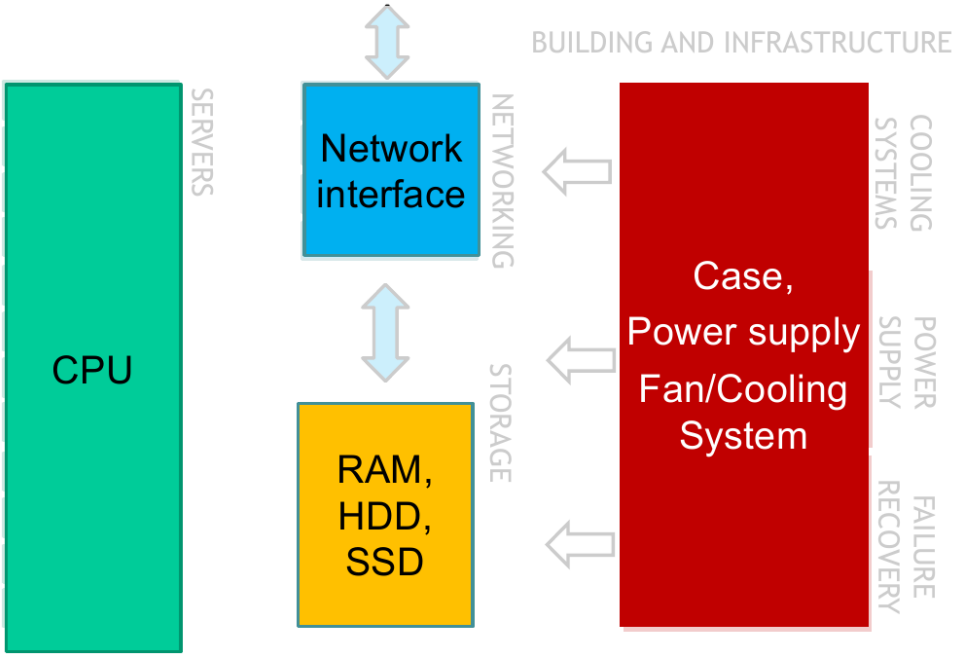

Figure 10: General Server Architecture

Servers can be considered as the basic building block of our DC, there exists several configurations and combinations of different types of servers (for bigger and smaller businesses) but all different assets are accomunated by the basic needs of a server in order to be called like that:

- **The Motherboard** - the central hub of the single server unit, connecting all components altoghether
- **Chipsets** - the main computational components, such as CPU, RAM, GPUs and main memory components.

Different types of servers are also to be considered:

- **Tower** - optimal for scalable solutions and easy cooling, but used only in businesses that require basic level of performance. Also consumes a lot of space and cable management is a nightmare.
- **Rack** - optimal for space usage and with racks (the shelves) an even better unified and interconnected system: unified because all is compared to units (1U = 44.45mm of height) to have the best rack organization. Their density often implies that they require more power to cool off and most importantly a certain difficulty in maintaining and replacing faulty ones.

- **Blade** - the latest type of servers, that function as a hybrid rack system. More compact, small, versatile and modular, meet the IEEE U standard for racks: they increase "packingness" at the cost of having a higher initial cost, possible vendor lock-in (no cables, only specific enclosures that are made by specific manufacturers) and a big need of cooling.

Figure 11: All Server Types

4.1 Accelerators and ML Era

The need of more complex model inside the industry led to new hardware to try to "accelerate" the all neural models and their computational load: let's define some basis of AI and Machine Learning to understand how those accelerators work

Artificial Neural Network

An ANN is a human inspired computational model where interconnected nodes (neurons) are organized in layers and are instructed to process and analyze data.

Single neurons are connected with weighted connections, that are strenghtened or weakened through training: initially, those are randomly initialized. That's how, with the advent of more powerful CPUs, Deep Learning started to take place. Since 2013, AI training compute requirements have doubled every 3.5 months, that's why accelerators were needed to speed up the training process of a model. The first solution that came to mind is the usage of GPUs, which have high compute density and are built for parallel operations with high throughput: more threads to work with and with deep pipelines with multiple stages.

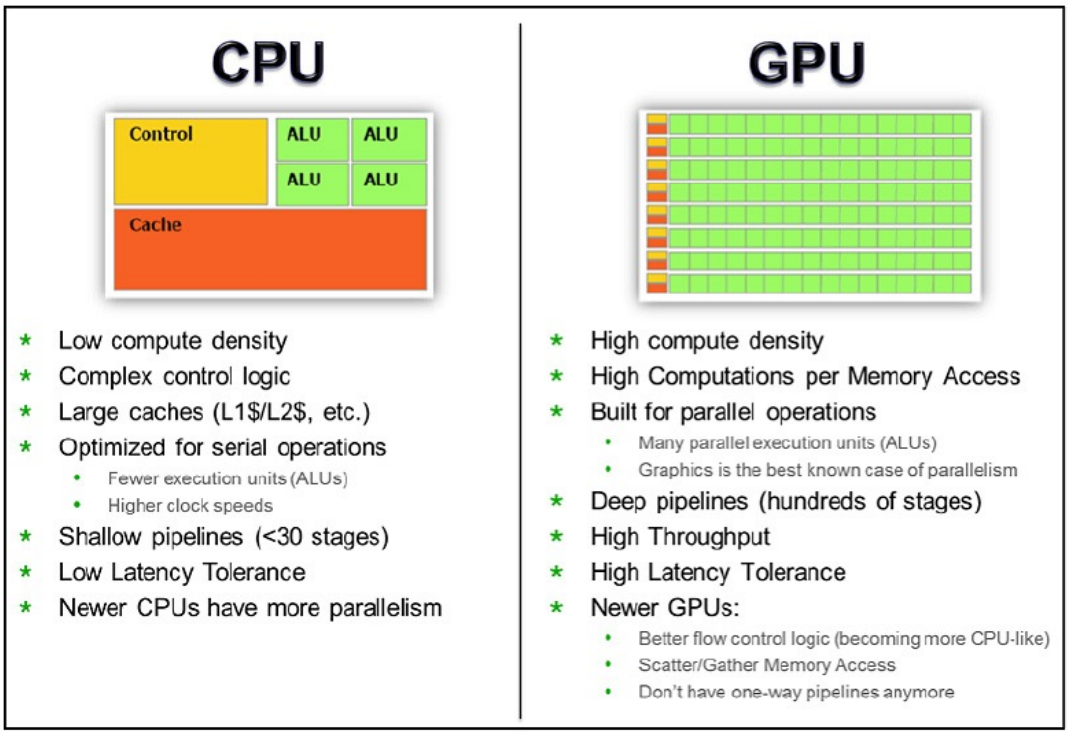

Figure 12: CPUs vs GPUs

Appropriate languages are to be used to program on these GPUs, such as CUDA. Once a network of GPUs is established, it is possible to determine the speed of the training by watching the slower GPU of the batch: every GPU does its calculations, then follow a back-propagation algorithm to update the weights on the connections and finally all informations are updated between all nodes. **It is crucial that the whole system is well connected.** Finally, the system is structured with this layout (as example): GPUs are connected with high-bandwidth connectors (such as NVlink) to minimize latency, a CPU hosts is connected to the accelerator with a PCI-express.

4.2 Tensor Processing Units

Since GPUs are still general purpose devices, Google decided to implement a device specialized in ML training: by having next to zero overhead efficiency is maximized.
Go on until page 45